Algebra Monsoon 2026 Lecture 8 : Adjoints, Representables

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Adjoint functors

Representable functors

Let $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$ be functors. An adjunction between F and G is a system of bijections

$$\varphi_{X,Y} : \mathrm{Hom}_{\mathcal{D}}(F(X), Y) \to \mathrm{Hom}_{\mathcal{C}}(X, G(Y))$$

indexed by pairs of objects X of $\mathcal{C}$ and Y of $\mathcal{D}$ such that for every $f \in \mathrm{Hom}_{\mathcal{C}}(X', X)$ and $g \in \mathrm{Hom}_{\mathcal{D}}(Y, Y')$ the diagram

$$\begin{array}{ccc}
\mathrm{Hom}_{\mathcal{D}}(F(X), Y) & \xrightarrow{\varphi_{X,Y}} & \mathrm{Hom}_{\mathcal{C}}(X, G(Y)) \\
\downarrow {\scriptstyle ( ) \circ F(f)} & & \downarrow {\scriptstyle ( ) \circ f} \\
\mathrm{Hom}_{\mathcal{D}}(F(X'), Y) & \xrightarrow{\varphi_{X',Y}} & \mathrm{Hom}_{\mathcal{C}}(X', G(Y)) \\
\downarrow {\scriptstyle g \circ ( )} & & \downarrow {\scriptstyle G(g) \circ ( )} \\
\mathrm{Hom}_{\mathcal{D}}(F(X'), Y') & \xrightarrow{\varphi_{X',Y'}} & \mathrm{Hom}_{\mathcal{C}}(X', G(Y'))
\end{array}$$

commutes.